# Underspecification restrictions in polyrhythmic processing(?)

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# Outline of the talk

- Structural underspecification as a core feature of Natural Language (NL)
  - Restrictions arising from the nature of underspecification
    - The one unfixed node at a time constraint
- Searching for underspecification restrictions in polyrhythmic processing
  - Interpreting the existing experimental evidence under the DS perspective
    - The need for new experimental evidence

# Person Restrictions - The Person Case Constraint

- Relevant only with clitics (the weak forms of object personal pronouns)
- Clitic co-occurence restriction
  - A dative clitic cannot co-occcur with a 1st/2nd person accusative clitic (usually dubbed as the Strong PCC)

(1) *Le me ha dado
it.CL-DAT me.CL-ACC has given
'He/She has given him to me.' [Spanish]

(2) *Paul me lui présentera
Paul me.CL-ACC him.CL-DAT will-introduce
'Paul will introduce me to him.' [French]

(3) *Mu se exi dosei
me.CL-DAT you.CL-ACC has given
'He/She/It has given you to me.' [Standard Modern Greek]

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- No semantic restriction is at play in these constructions
  - Substituting one of the clitics of an illicit cluster with a strong pronoun renders the sentence grammmatical

(8) Me sistisan se sena
me.CL-ACC introduced to you
'They introduced you to me.' [Standard Modern Greek]

(9) Je t' ai presenté à lui
I you.CL have introduced to him
'I introduced you to him.' [French]

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- 3rd person accusative clitics (e.g. Spanish lo, Greek to) as inducing fixed structure (the sole interpretation is the direct object one)

# The DS view - Dative and 1st/2nd person clitics

- Dative and 1st/2nd person clitics as projecting unfixed nodes
  - Locally unfixed to capture the locality restrictions associated with clitics

| | | |
|---|---|---|
| IF | $?Ty(t)$ | |
| THEN | IF | $[\downarrow_1^+]?Ty(x)$ |
| | OR | |
| | IF | $Mood(Imp)$ |
| | THEN | $\mathtt{make}(\langle\downarrow_1^*\rangle); \mathtt{go}(\langle\downarrow_1^*\rangle);$ |
| | | $\mathtt{make}(\langle\downarrow_0\rangle); \mathtt{go}(\langle\downarrow_0\rangle)$ |
| | | $\mathtt{put}(\langle\uparrow_0\rangle\langle\uparrow_1^*\rangle?Ty(t));$ |
| | | $\mathtt{make}(\langle\downarrow_0\rangle); \mathtt{go}(\langle\downarrow_0\rangle)$ |
| | | $\mathtt{put}(Ty(e), Fo(U_{x'}), ?\exists\mathbf{x}.Fo(\mathbf{x}))$ |
| | | $\mathtt{put}(?\exists\mathbf{x}.Tn(\mathbf{x}), gofirst(?Ty(t)))$ |
| ELSE | Abort | |

# The DS view - 3rd person accusative clitics

- Projection of fixed structure

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No unfixed node relation

# The DS account - The PCC in Spanish/Italian

- Given these lexical entries the PCC is explained via a hard-wired logic constraint that bans the projection of more than one unfixed node
- PCC constructions will involve the projection of two locally unfixed nodes, something not allowed by the restriction on underspecified relations

After parsing *me* in *me-te*:

$$Tn(a), \ldots ?Ty(t), \Diamond$$

$$\langle \uparrow_1^* \rangle Tn(a), ?\mathrm{Ty(x)}$$

$$\langle \uparrow_0 \rangle \langle \uparrow_1^* \rangle Tn(a),$$
$$Ty(e), Fo(U_{Speaker'})$$
$$?\exists \mathbf{x}.Fo(\mathbf{x}), ?\exists \mathbf{x}.Tn(\mathbf{x}).$$

# One unfixed node at a time

After parsing *te* in *me-te*:

$$Tn(a), ...?Ty(t), \Diamond$$

$$\langle\uparrow_1^*\rangle Tn(a), ?\mathrm{Ty}(\mathrm{x})$$

$$\langle\uparrow_0\rangle\langle\uparrow_1^*\rangle Tn(a),$$
$$Ty(e), Fo(U_{Speaker'}), Fo(U_{Hearer'})$$
$$?\exists\mathbf{x}.Fo(\mathbf{x}), ?\exists\mathbf{x}.Tn(\mathbf{x}).$$

- The two nodes collapse into one, carrying incompatible Formula values. The parse cannot be successful.

# Music and language processing

- Does language and music share a common structure building mechanism? If the answer is yes, then general structure building mechanisms should be common to both.
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- Does language and music share a common structure building mechanism? If the answer is yes, then general structure building mechanisms should be common to both.
  - Is the notion of structural underspecification as a structure building mechanism relevant for polyrhythmic processing?
    - In what follows,I present some preliminary thoughts (rather naive) on how such a restriction can be traced when looking at polythythm processing

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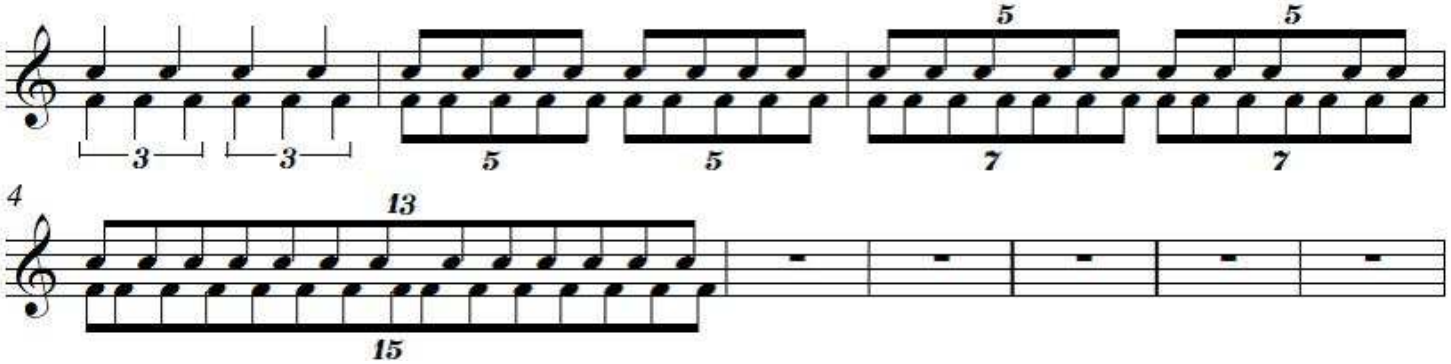

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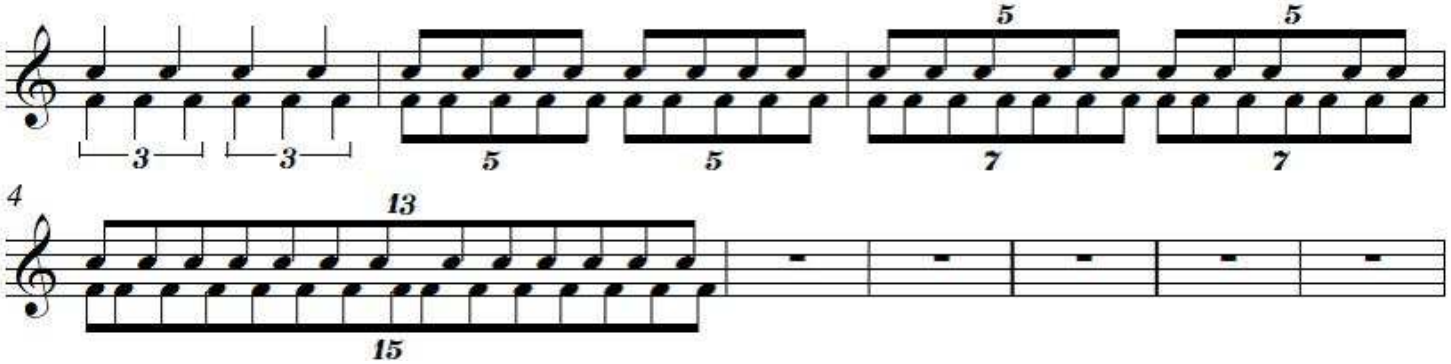


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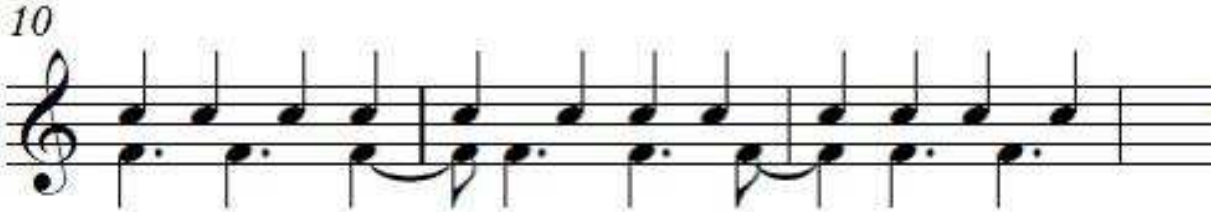

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     - Relevant factors include presentation rate (tempo), frequency separation (the pitch difference of the two rhythms) and attentional behaviour (see for example Jones 1976; Beauvillain, 1983; Bregman, 1990; Moealants, 2005 and Fidali et al., 2011 for a summary)

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     - Various studies suggest that the key to successful production of polyrhytms is integration (Summers & Magill, 1996; Summers, 2002 inter alios)

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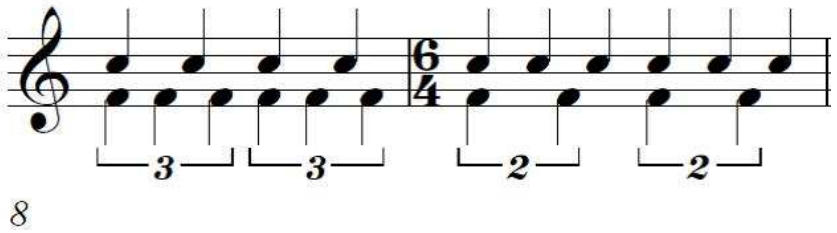

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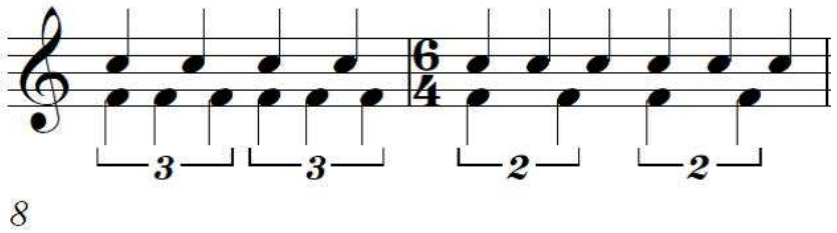

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  - A well-known fact among musicians, especially those dealing with polyrhythms (e.g. drummers, pianists)
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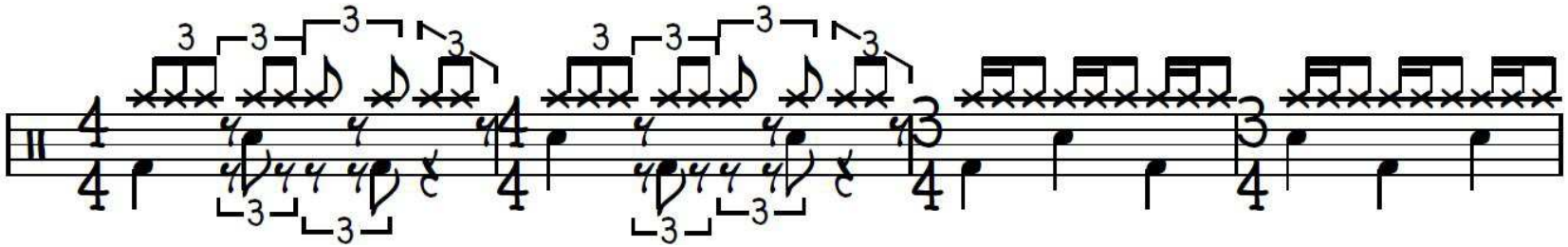

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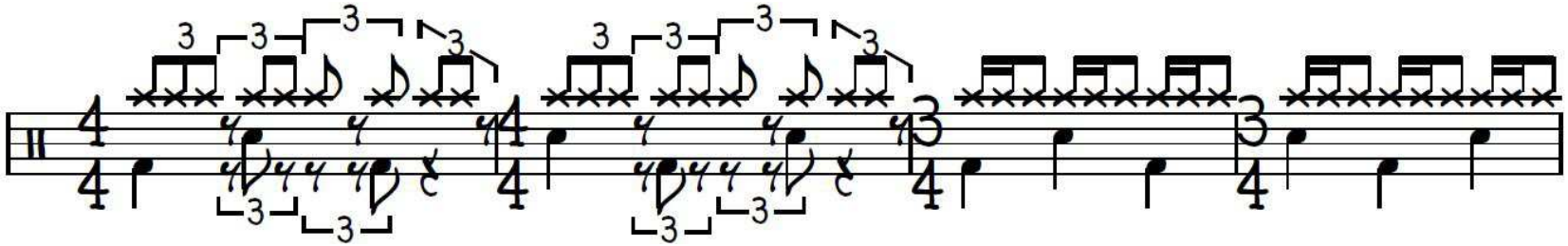

  - However, most of the drummers will find the rhythm in the first two bars harder to play than the rhythm in the third and fourth or some will manage to play the latter only

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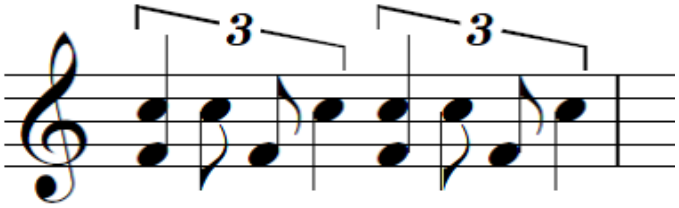

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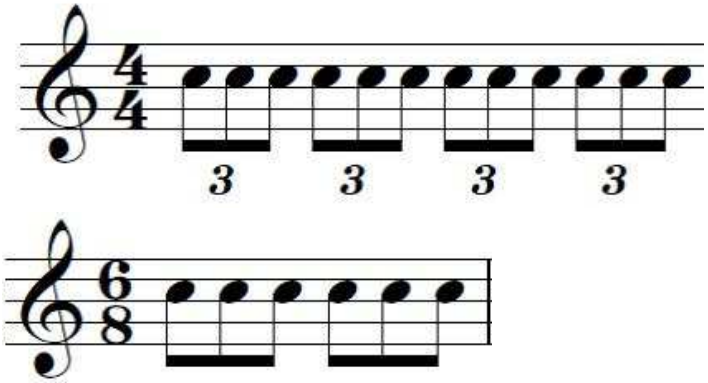
    Some of these interpretations



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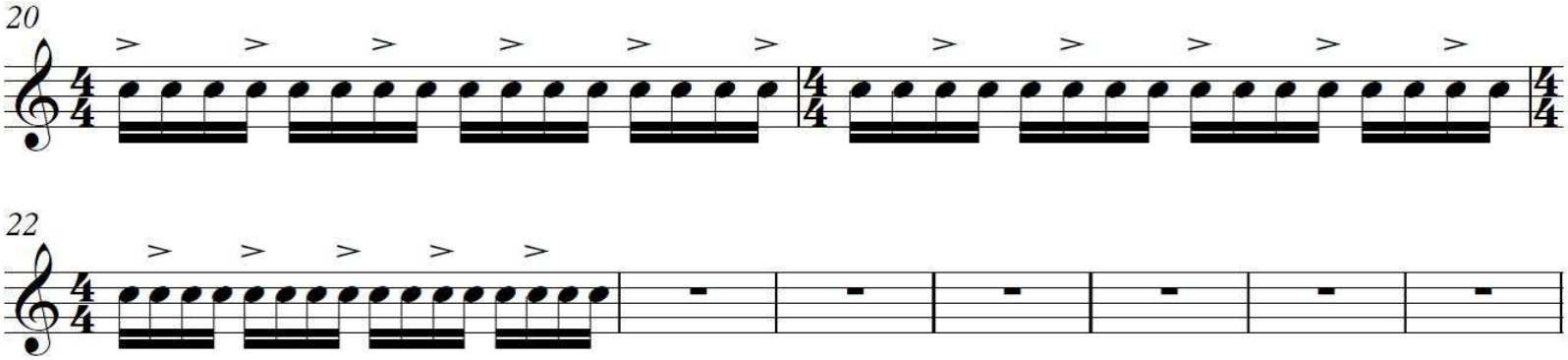

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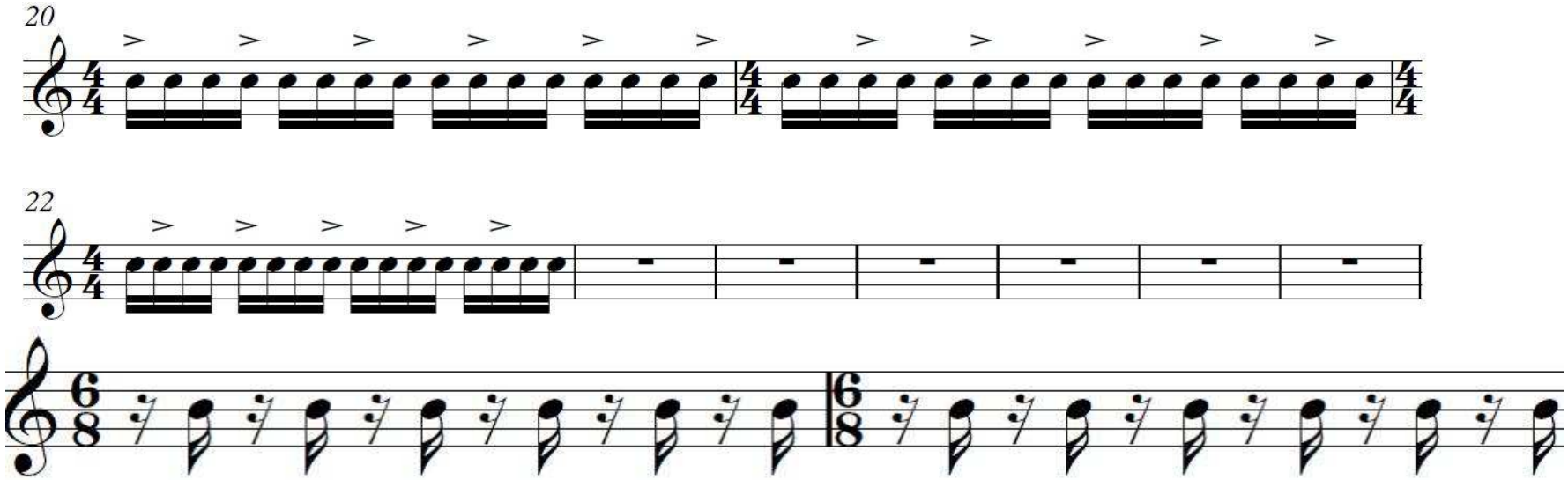

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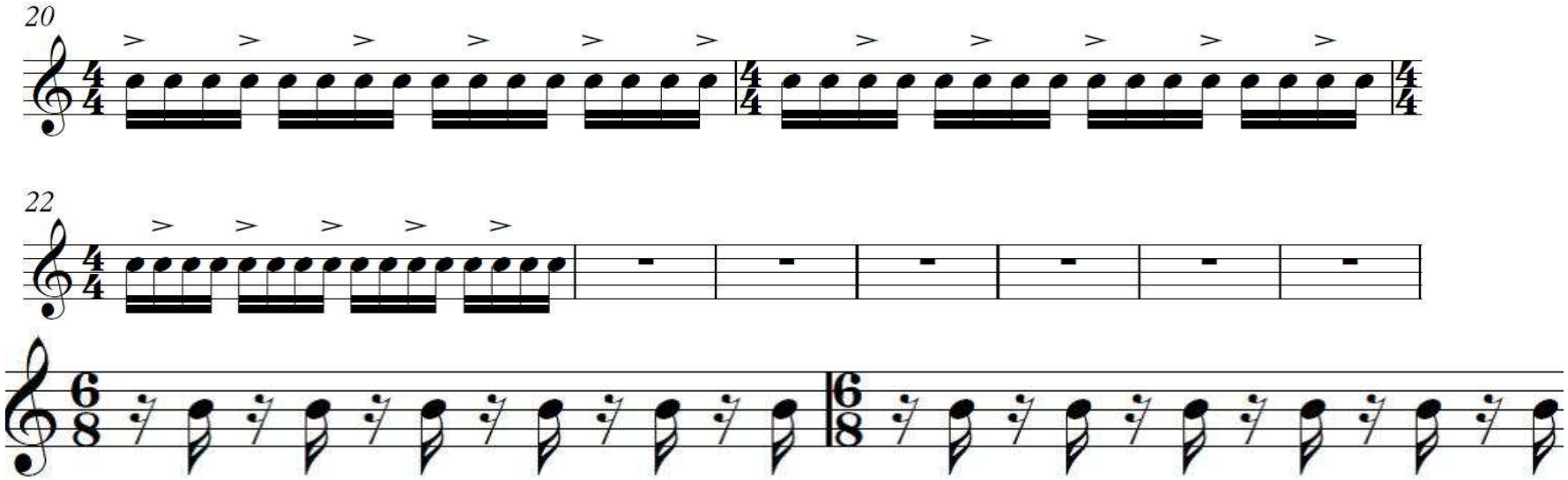

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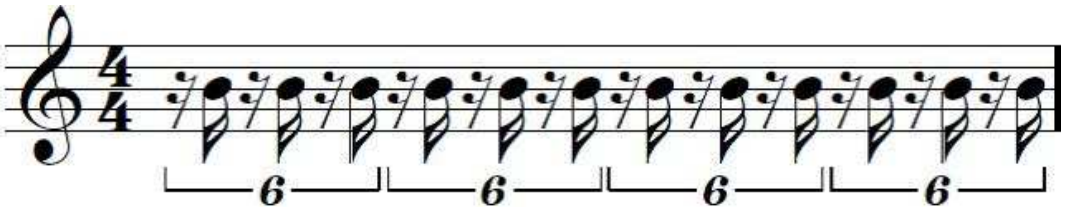

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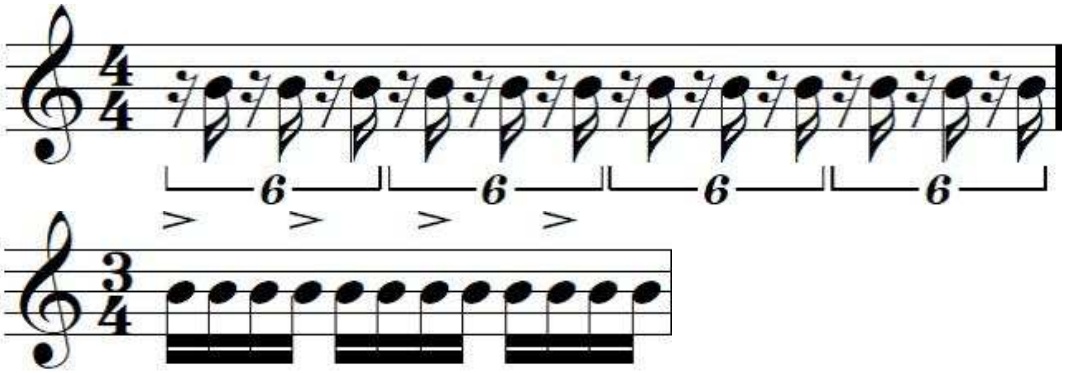

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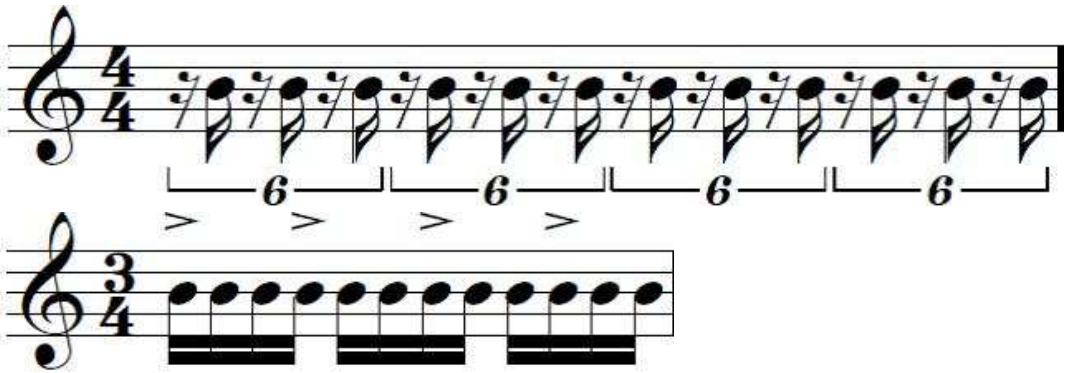

Of course, many listeners will fix their interpretation to one of the two most plausible interpretations (triplet in 4/4 or 6/8). What makes fixing possible here is another issue (see London, 2004 for a discussion). In principle however, such rhythmic units when played out of metrical context can be underspecified

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- The issue arises when prior to the fixing of the underspecified relation we have another underspecifed rhythm at play

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  - In effect we do have a polyrhythm going with no established metrical context.

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  a. Extract a composite pattern out of all the streams and then match it to a suitable metric framework
  b. Focus on one stream while treating the other as 'noise' (London, 2004)
- The question is what happens when no such metric context has been established? In this case, in both options, the processed rhythms are underspecified with respect to a not-yet-established metrical context

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    - What does ungrammaticality mean in terms of parsing polyrhythms? Alternatively, what is the equivalent effect in polyrhythmic processing?
    - If this restriction has anything to offer in terms of music processing then we should have an idea of what does this restriction amounts to (for example can we say that failure to tap to any of the individual rhythms or to tap to only one of them is the result of such kind of restriction?)

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  - It is however unclear how one can check failure of parsing a polyrhythm since the inability to tap along might be due to motor limitations and not perceptual ones
  - Experimental evidence have shown that the case of segregation where both streams are processed is not possible. If both these streams are underspecified, then this unavailability could be the result of the constraint
    - However, we cannot be sure that in all cases both rhythms are underspecified
  - Thus, one cannot be conclusive at least on the basis of the existent data on polyrhythmic processing

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  - However, we must be cautious here: compatibility with the data and the existence of the constraint are too separate issues
    - The former does not necessarily imply the latter
  - More evidence is needed in order to determine whether such a constraint is indeed operative in polyrhythmic processing
  - We need experiments designed specifically for this purpose

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    - Furthermore, correlation between polyrhythmic processing and specific aspects of language processing has been established (activation of Broadmann area 47 in polyrhythmic processing, Vuust et al, 2011)
- However, this is for the moment all speculative (and potentially naive) and nothing can be said with more confidence until empirical evidence further backs such claims